

# Beyond the Model

## The critical role of data filtering in clinical machine learning

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Filtering is a modelling choice. Count what you removed.

0 patients removed by the seven value-level filters

44,663 of 46,032 stays removed by one record-level filter

38% → 61% prolonged-stay prevalence among its survivors

1 Filtering is a modelling choice

Clinical ML papers open with a filtering pipeline and rarely report what it removed. Any rule that edits or discards values changes the distribution, so the question is not *did filtering help?* but *is it still the same problem?*

- It can make a benchmark easier without making a model better.
- It is not task-neutral: one filter can help one task and hurt another.
- Pipelines standardise it; almost none characterise what it did.

2 One cohort, one model, 13 arms

MIMIC-III  
46,032 ICU stays

first 24 h  
7 vitals × 24 hours

VALUE filters  
(pre-aggregation)

random forest  
300 trees

FEATURE / RECORD filters  
(post-agg.)

hourly mean  
→ 168 cells

5-fold × 4 repeats · folds fixed once on the raw cohort by admission id threshold by Youden's J on an inner split → compare every arm with raw

Two tasks on the same 168 features: **prolonged ICU stay** (> 3 days; 38% positive) and **in-hospital mortality** (10%). Each filter is applied alone to the raw data, and every arm inherits the same folds, so arms are paired patient by patient.

3 Eleven filters, three levels

VALUE

FEATURE

RECORD

VALUE

FEATURE

RECORD

RECORD

RECORD

Only a record-level filter can delete a patient.

4 The raw data really is broken

The heart-rate column peaks at **9,999,999** bpm, a charting sentinel rather than a patient. Removing 507 of 1.16 million readings (0.04%) takes the maximum to 459 and the standard deviation from 9,291 to 24.

heart rate

systolic BP

diastolic BP

mean BP

respiration rate

temperature

oxygen saturation

Implausible readings removed per vital by the value-level filters.

Filtering is necessary. The question is what else it does.

5 Filtering moves the centroids

Centroid shift per vital after the all-vitals filter, by outcome class (paper Fig. 3).

Outliers are one-sided, so removing them moves the mean. Read across one vital's bars: if the classes move by different amounts, the filter treated them differently.

6 ...and it changes task difficulty

Same features, folds and model; only the filter changes. Accuracy moves both ways, often in opposite directions for the two tasks. The record-level filters move it most, for a reason shown in §10.

7 Did the predictions change?

McNemar's paired test ignores agreements and asks whether the filter fixed and broke equally many patients. All arms use the baseline's threshold. No per-vital filter is significant (all  $p \geq 0.06$ ).

8 How many patients survive?

10 value & feature arms

long missing segment

long gap

high invalid data

The value-level filters edit 21,930 of 6.69 million readings (0.33%) and remove no one. *High invalid data* sounds lenient, tolerating 16 blank cells in 168, yet keeps **1,369** of **46,032**.

9 The survivors are not random

Record filters keep densely monitored stays, and ICU monitoring tracks acuity: missingness is informative. So they select sicker patients. Prolonged stay goes from a 38% minority to a **61% majority**.

10 The bars compare different people

The *unchanged* baseline already scores worse on each filter's survivors (ICU, high invalid data: 0.698 → 0.625). That is -0.073 of the published -0.078. On the same 1,369 stays at one threshold, the filter itself moves accuracy by -0.007.

11 McNemar only sees survivors

Deleted patients never enter the test. High invalid data shifts the ICU base rate by 23 points, yet  $p = 0.515$ : silence here is lost power, not a clean bill of health.

12 Three questions, asked together

all vitals

high invalid data

Row one is artifact removal; row two is a different problem. The paper published the first two columns. The third costs one call to `len()`, and it explains the other two.

13 Take-aways

- Filtering is part of the method, not housekeeping.
- Level matters more than threshold. 0.33% of readings edited, no one removed, versus 97% of stays gone.
- Report the surviving  $n$  and its prevalence beside every metric, and score the baseline on the same survivors.